

A Review on Image Dehazing using Generative Adversarial Network (GAN)

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Abstract— The revolution in computer vision techniques and deep learning has played an important role. Among all challenges faced in this field, image dehazing remains a tricky problem. Haze can be caused by adverse or foggy weather conditions, unsuitable lighting, and pollution. The problem increases in intensity, especially in extreme conditions. A deep learning-based approach was developed to solve problems. But learning-based methods still face some problems. They are poor at taking out haze from outdoor images with variable training data due to insufficient use of a variety of training data. This paper presents a series of dehazing techniques, and explores, in particular the GAN-based approach. These GAN based approaches have been introduced to resolve the issues facing image dehazing.

GANs are trained using a large collection of haze free and hazy images for the removal of haze from degraded ones and improve computer vision performance in hazy situations. GAN-based image dehazing is widely accepted as the best method for real time dehazing. The present paper qualitatively analyzes different approaches and discusses their strengths and weaknesses. In addition, this review discusses the mainstream benchmarks, existing challenges, performance metrics, and applications in image dehazing research.

Index Terms—Atmospheric haze, Generative Adversarial Network (GAN), Dehazing, Performance evaluation metrics.

I. INTRODUCTION

Haze has a major impact on machine vision systems' performance. Haze affects images by light attenuation and scattering phenomena. Due to reducing scene visibility, which means that it reduces performance in various tasks such as detecting objects, tracking targets and segmenting the scene, so image dehazing is widely considered a key area of machine vision. Image dehazing enhances overall performance by lowering noise, sharpening edges, and increasing the signal-to-noise ratio. This is essential in domains such as meteorological research, geographic information research, security monitoring, and assisted driving.

Advances in image processing technology have also developed an approach to image dehazing. In earlier, image dehazing was considered an image enhancement problem, where techniques like retinex and contrast enhancement were applied for defogging. The researchers then used image restoration methods based on ASM to resolve dehazing approaches, including DCP [1], NLD [2], Haze line prior [3], and color attenuation prior [4]. Fusion-based approaches to improve performance by combining multiple methods. In earlier, the fast development of image dehazing deep learning algorithms like MSCNN[5], DehazeNet[6],

FFA-Net[7], DCPDN[8], FAMED-Net[9] and AOD-Net[10] is improved in effect due to image dehazing and robustness. GAN-based dehazing methods [11, 12, 14, 15], FD-GAN [16], prior-based with adversarial learning-based dehazing techniques [17], Self-attention GAN [13], DehazeGAN [18], and CycleDehaze[19] are implemented, which obliterate haze from hazy images without introducing artifacts or color saturations.

A. The Taxonomy of Image Dehazing methods

1. The first dehazing technique was image enhancement, which was efficient and clear but was inflexible, resulting in very bad performance in complex scenes.
2. The image restoration method is using atmospheric scattering model (ASM). The prior knowledge is leveraged to transform hazy images into clear ones. Many of the methods provide good quality results at low computational cost.
3. Fusion based methods are well-known for their robustness and speed over deep learning-based methods, which create more than one derivative image from a hazy image and extract the features with fusion for dehazing.
4. Dehazing techniques using deep learning rely on learning the transformation from hazy to clear images using datasets of realistic fog-free images, extracting features, and categorizing based on ASM, which improves performance and robustness.

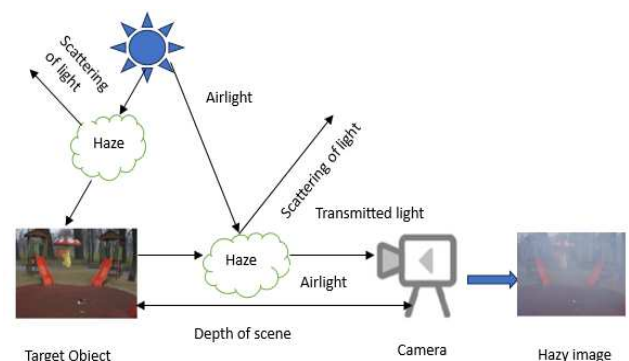


Fig.1. Atmospheric scattering Physical model

Haze in ASM occurs by tiny particles in the atmosphere scattering light, which reduces contrast and field of view. This phenomenon, first proposed by Koschmieder [44] in 1925 and later by McCartney et al. [45] in 1976, is attributed to atmospheric scattered light.

In 1999, Narasimhan [20] and his colleagues created a mathematical model as shown in Fig. 1 to explain how foggy images are formed, highlighting the main reasons for degradation in detection systems under strong scattering

mediums. The model explains that background light intensity often exceeds target light intensity, is usually used to simulate the degradation process of hazy images.

B. Atmosphere Scattering Model (ASM)

The ASM formula is as follows:

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad (1)$$

The dehazing method uses $t(x)$ called the transmission medium, where the position of pixels is represented by x , A represents global atmospheric light, $I(x)$ represents hazy image, and $J(x)$ represents haze-free (clear) image.

The Transmission medium $t(x)$ is represented as

$$t(x) = e^{-\beta d(x)} \quad (2)$$

The clear image $j(x)$ can be achieved by calculating the β and $d(x)$, where β and $d(x)$ represents atmosphere scattering parameter and the scene depth, which is essential for dataset synthesis. $t(x)$ means the transmission medium.

The $j(x)$ can be calculated as

$$J(x) = \frac{I(x) - A(1 - t(x))}{t(x)} \quad (3)$$

The paper provides an overview of Generative Adversarial Networks in Section 2, Section 3 discusses a general outline of image dehazing techniques, and Section 4 involves a review of GAN-based dehazing methods. Section 5 and 6 discuss benchmarks and performance evaluation metrics, and Section 7 presents a comparative analysis of different GAN-based dehazing techniques based on features and limitations. Section 8 discusses real-world applications. Finally, the last section provides the conclusion of a study.

II. GENERATIVE ADVERSARIAL NETWORK

Recent advancements have highlighted the impressive performance of GANs [21]. GANs have been commonly applied across various domains as a powerful deep generative model. This section discusses how GANs are created and how they are trained.

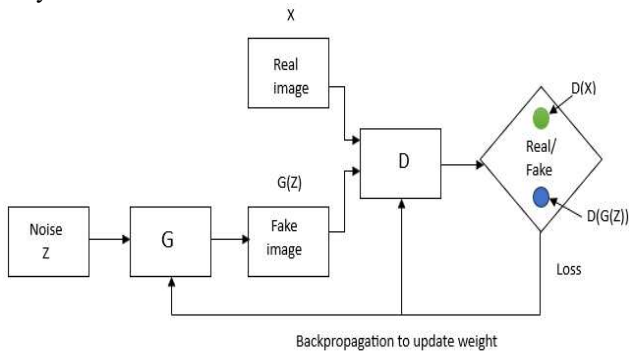


Fig.2 The Generative Adversarial Networks Architecture

A. Basic Theory

The GAN architecture [22] as shown in Fig. 2 consists of the structure of two layers Generator G and Discriminator D . An input noise (z) that is drawn from either a Gaussian or a uniform distribution is transformed into a new distribution to generate samples which is synthetic $G(z)$ from it. Simultaneously, the discriminator D has two signal inputs: $G(z)$ the generated samples generated from generator G , and the real images (x) of the actual data set instead of a generated one. The discriminator is also a binary classifier that measures the likelihood that the said sample actually

originated from the real data set instead of from a generated one. The discriminator (D) finds it more and more difficult to discern between created and actual samples as the generator (G) is trained to generate samples that closely resemble the real data distribution, in accordance with the rules of game theory. On the other hand, the discriminator is tuned to correctly identify inputs as synthetic or real. The classification accuracy in relation to actual data drives this adversarial process, which guarantees that both networks gradually enhance their performance. While the generator successfully learns the underlying distribution of the real dataset, the discriminator eventually loses its ability to distinguish between created and genuine data.

B. Training Mechanism

Throughout the training procedure, G with D are iteratively updated together to improve both networks performance[22]. The generator's (G) objective function is given as a minimax game, as illustrated in Equation (4).

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(X)} [\log D(x)] +$$

$$E_{z \sim p_z(Z)} [\log (1 - D(G(z)))],$$

(4) In this case, x represents actual data points sampled from the distribution $p_{data}(X)$, while E represents the expected values. The vector z is generated from a random noise distribution $p_z(Z)$. The terms $G(Z)$ represent the output samples of the generator. $D(X)$ represents the probability that the discriminator correctly identifies x as real sample. Similarly, $D(G(Z))$ represents likelihood that discriminator correctly identifies the generated samples as real. To deceive the discriminator, the generator G aims to maximize $D(G(Z))$, which is equivalent to minimizing the logarithm of $1 - D(G(Z))$. Discriminator D , employs a pass-entropy function to differentiate between real data x and generated information $G(Z)$, enabling you to optimize the characteristic $V(D, G)$. In practice, the generator is kept constant. The discriminator is trained to accurately differentiate between real and generated images. After a completion of optimization process, the system saves changes consistently, while the generator is trained to improve its overall performance.

GANs, a key innovation, generate data in complex domains and scenarios, expanding datasets for machine learning applications. Their realistic images make them popular among researchers and developers.

III. TECHNIQUES OF IMAGE DEHAZING

There are different image dehazing methods typically depends on robust priors or underlying assumptions.

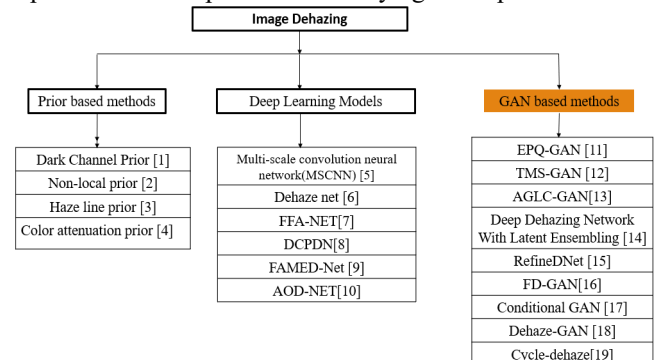


Fig.3 Classification of Image Dehazing Methods

Fig.3 illustrates the classification of image dehazing methods contains prior-based, deep learning-based, and GAN-based approaches.

A. PRIOR-BASED OR STATISTICAL METHOD

The statistical dehazing techniques relies on utilizing robust priors or assumptions. The primary priors commonly used in image dehazing are including the following: **Dark channel prior (DCP)**: In their studies, the author [1] introduced a unique assumption primarily based on haze-free outside photos, highlighting the presence of low-intensity pixels in non-sky regions. These dark pixels, normally better in hazy images, indicate haze thickness and create a transmission map. However, it has limitations like halo effects, slow matting refinement, and difficulty distinguishing gadgets and backgrounds. **Haze-line**: Earlier, Berman et al. [3] delivered the haze-line previously (NLD), which approximates colorings in clear pix the use of wonderful color agencies in RGB area. This fast, linear approach handles exceptional sizes and patch sizes without schooling. However, it relies on statistical assumptions about shade distribution, probably now not relevant to unnatural scenes. **Color Attenuation Prior (CAP)**: Zhu et al. [4] in earlier work introduced CAP, it is found that in hazy image shade saturation and brightness of pixels were determined that drastically changes with the versions in haze awareness, leading to the development of the CAP. Although CAP is fast and efficient technique however it faces challenges in managing homogeneous atmospheric situations.

B. DEHAZING APPROACHES LEVERAGING DEEP LEARNING

The Dehazing approaches leveraging deep learning models had revolutionized by automatically extracting relevant features, a significant improvement from traditional methods [1,3,4]. This has led to the creation of novel techniques to remove image haze effectively. In the past few decades, deep learning-based methods emerged as highly effective alternatives to traditional prior-based dehazing techniques. The goal of deep learning techniques is to create transmission maps and estimating atmospheric light for getting clear images, unlike previous methods that relied on prior knowledge such as MSCNN[5], DEHAZE-NET[6], FFA-NET[7], DCPDN[8], FAMED-NET[9], AOD-NET[10].

IV. GENERATIVE ADVERSARIAL NETWORKS (GANS) FOR DEHAZING

GANs were first introduced by Goodfellow et al. [21]. Researchers increasingly utilize GANs for removing haze in images also in image denoising, image classification, and translation. In the past few years, several innovative dehazing models have been introduced, significantly bridging the gap dehazing images from hazy one with impressive results.

Keshampeta Ashwini et al., [11] introduced the Evolutionary Perceptual Quality Assessment Generative Adversarial Network (EPQ-GAN) to resolve the issue of dual GAN architecture of generator and discriminator. Inspired by E-GAN, EPQ-GAN leverages the DE algorithm (Differential Evolution) to optimize the training of the generator network, resulting in more efficient haze removal. U-Net architecture is adopted for generator network, and

discriminator acts as a binary classifier. Method reliance on the paired dataset and low PSNR value for the real-world dataset.

Pengyu Wang et al., [12] To address domain shift challenges, author developed the Twofold Multi-scale GAN in single-image dehazing. This framework comprises two dedicated modules: HgGAN means Haze-generation GAN that applies realistic atmospheric distortions to synthetic inputs, and a HrGAN means Haze-removal GAN structured for elimination of haze across both artificial and authentic imagery. Cooperative training mechanisms facilitate superior real-world adaptation. Notably, HrGAN's dual-branch architecture leverages haze residual mapping, enhancing structural integrity while optimizing color fidelity and detail preservation. A breakthrough in synthetic-to-real adaptation an integrated multi-attention progressive fusion module (MAPFM) further enables precise feature integration across perceptual levels. Due to complex network of dual-GAN architectures failure cases still occur when recovery is challenging due to blurry images that are too dim.

Hongyuan Zhu et al.[18] introduced DehazeGAN, which is among the first to integrate image dehazing with Generative Adversarial Networks (GANs). This work redefines the ASM within a new GAN by taking inspiration from differentiable programming. The HazeCOCO dataset is introduced to support this work, which contains 0.7 million synthetic outdoor and indoor images. This work is limited to the outdoor dataset, images are inaccurate.

Engin et al.[19] Enhanced CycleGAN, known as CycleDehaze, is a method proposed by author that combines perceptual and cycle consistency losses to remain the visual texture of the image, which is dehazed. This method is characterized by two discriminators, represented as D_x and D_y , along with two generators, represented as G and F . Cyclic Perceptual-Consistency Loss function ensures that image details and structures are retained by using an integration of high and low-level features derived from the pooling layers of the VGG16 network. The dual GAN method uses perceptual loss to train the generator, than pixel space loss. But limitations are decreased image sharpness in outdoor environments.

Yan Zhao Su et al., [17] Yan Zhao Su and colleagues recently brought out a new twist with their prior image-guided cGAN for single-image dehazing. Instead of sticking to a cookie cutter approach, this model grabs helpful cues from a prior dehazed image using an attention-based feature mixer that digs out important details. Additionally, it employs an adaptive group norm technique during decoding, which, to a certain degree, facilitates a style transfer that noticeably enhances the appearance of the reconstructed image. In most instances, the final dehazed images typically exhibit fewer color errors, primarily due to the guidance offered by DCP.

Dong et al., [16] introduced GAN-based an innovative dehazing method. The generator is guided by a fusion discriminator in this method to produce images that closely resemble real ones. The overall performance is improved when a training dataset is introduced. They present a novel model that incorporates two important features: to help the generator fusion discriminator is introduced, incorporating additional assumptions based on the characteristics of high

and low frequencies. This improves the discriminator's capability to assess the variations in color and texture between hazy images and dehazed images. Structural information, like smooth areas, is captured by using a Gaussian filter, while the Laplacian operator highlights high-frequency features like fine details and edges.

Qu et al. [23] introduced the improved pix2pix generator. (EPDN), specifically designed for image dehazing with advanced algorithms highlighting of subtle aspects. The framework contains three main elements: a generator that starts with a high resolution and progressively down samples it. Methodology, employing two distinct sub-generators, the first sub-generator processes the image at a low resolution, while the second refines it at a finer scale. The model incorporates two discriminators—one for images and one for text. However, EPDN exhibits certain limitations, such as a tendency to produce over-smoothed results in regions with complex textures or dense haze. **Zhang et al. [24]** suggested the HDDNET hierarchical density-aware Dehazing network created to improve image quality. Dehazing is a technique that improves the clarity of images by removing the haze layer that is caused by atmospheric scattering of light. The model comprises three essential elements: a pyramidal encoder for Hazy Image Processing, and transforming into feature vectors. Unlike conventional encoders, models are able to encode data in a more efficient and effective manner. This design merges downscale attributes from previous layers into subsequent layers. This model has multiscale features and captures global information more effectively.

Yang et al. [25] introduced a disentangled Dehaze Network to produce visually appealing dehazed images rather than strictly restoring the original image. Their approach disentangles hazy images into three key components: global atmospheric light, scene radiance, and transmission map, each generated by separate networks. Furthermore, while the method achieves improved visual quality, it may still struggle with fine texture recovery and color fidelity in highly degraded regions.

V. BENCHMARKS

There are challenges with image dehazing, especially when attempting to capture a pair of clear and hazy images. As technology has advanced, various stages of acquisition have been adopted for image dehazing datasets, including virtual scene generation, synthetic data creation, and real data collection, as indicated in Table II.

Table II. A Comparative Study of Datasets for Image Dehazing

Type	Dataset	Year	Feature
Virtual	FRIDA [28]	2010	Outdoor scenes with various levels of haze densities of paired hazy images and clear images
Virtual	FRIDA [28]	2012	Paired outdoor hazy and clear images of varying haze densities
Virtual	FROSI [29]	2014	Road sign identification in Outdoor Settings images
Synthetic	NYU Depth-V2 [30]	2012	RGB and depth images in indoor environments
Synthetic	D-Hazy [31]	2016	Real-world paired images rather than

			software generated ones
Synthetic	HazeRD [32]	2017	Paired outside hazy and haze-free images
Synthetic	RESIDE [33]	2017	Large-scale and most widely used paired images both outdoor and indoor
Real-World	I-Haze [34]	2018	Paired Indoor Hazy and Clear Real-World Images
Real-World	O-Haze [35]	2018	Paired outdoor Hazy and Clear Real-World Images
Real-World	Dense-Haze [36]	2019	Densely paired real-world outdoor images containing both hazy and corresponding haze free versions
Real-World	NH-Haze[37]	2020	Paired outdoor images featuring non homogeneous haze and their corresponding clear versions

VI. PERFORMANCE METRICS AND EVALUATION

Along with dehazing methods and benchmarks, performance evaluation metrics are essential in image dehazing research, as they provide a consistent framework for evaluating and comparing the efficiency and quality of various dehazing techniques.

i. Peak Signal-to-Noise Ratio (PSNR)

PSNR [38], a popular full-reference image quality assessment metric, compares the restored image to the ground truth to determine the degree of reconstruction accuracy. Its value is measured in decibels (dB). It is calculated using mean squared error (MSE).

$$MSE = \frac{1}{RXC} \sum_{n=1}^R \sum_{m=1}^C [Gt(i, j) - Or(i, j)]^2 \quad (5)$$

Gt(i, j) denotes the pixel intensity at position (i, j) in the ground truth image, while Or(i, j) represents the corresponding pixel intensity in the restored image. R and C are the number of rows and columns. The variables m and n are the coordinates of individual pixels

A higher value of PSNR indicates better image quality and less distortion, where a lower MSE results in a higher PSNR, signifying a closer resemblance to the original image.

It is calculated as:

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (6)$$

MAX is 255, which represents the maximum intensity value of pixel color in an 8-bit image. Dehazing methods PSNR effectively measure image clarity.

i. Structural similarity index metric (SSIM)

It [39] is a standard full-reference image quality assessment metric. It assesses a resemblance between a hazed image and a clear one. SSIM's value falls between -1 and 1. Unlike PSNR and MSE, SSIM focuses on pixel-wise differences. SSIM's higher value shows that the dehazed image will more closely resemble the corresponding clean image, hence better performing dehazing itself. Where higher SSIM shows better perceptual similarity and higher image quality, it serves great utility in image dehazing and captures structurally distorted visual quality improvements.

VII . COMPARATIVE ANALYSIS

Table III presents a detailed analysis of key image dehazing techniques introduced recently. The analysis considers

several parameters, including method, technique, key findings, limitations and evaluation metrics of each proposed method.

Table III A Comparative analysis of different Image dehazing techniques

Sr. no	Year, author, and Publication	Method	Technique	Feature	Limitations	Metrics
1	2024, Keshampeta Ashwini et al., IEEE Transactions[11]	EPQ-GAN	Differential evolution (DE) algorithm	Dehaze images without any artifact and color saturations	Reliance on the paired dataset and low PSNR value of the real-world dataset	Frida-2 dataset PSNR:31.27 dB SSIM:0.9687
2	2022, Pengyu Wang et. al., IEEE Transactions [12]	TMS-GAN	Twofold Multi-scale GAN	Progressively fuse multi-scale features due to Multi-attention mechanism. Adaptive real-world dehazing performance	Method is robust but nighttime & dim hazy images fail to recover	SOTS dataset PSNR:35.54dB SSIM:0.990
3	2023, Jaisurya et.al.,Elsevier[13]	AGLC GAN	Attention based Global Local Cycle consistent GAN	Unpaired dehazing methods along with the feature attention mechanism	Image fine details may be lost or smoothed out to satisfy the cycle-consistency requirement.	RESIDE ITS PSNR:32.30dB SSIM:0.9706 RESIDE OTS PSNR:36.93dB SSIM:0.9816
4	2021,Li et.al., IEEE Transactions[14]	T-Net and D-Net	Adversarial learning	Adversarial learning-based photo-realistic dehazing method & optimized for high-level vision tasks	Sequential and parallel configurations, with adversarial learning is computationally expensive	O-Haze Dataset PSNR:24.34dB SSIM:0.7494
5	2021, ZHAO et al.,IEEE Transactions[15]	RefineDNet	prior-based DCP and learning-based	DCP via adversarial learning using unpaired hazy and clear images	Some regions of dehazed images are oversaturated	SOTS Dataset PSNR:24.23dB SSIM:0.9431

VIII. APPLICATIONS OF GAN-BASED IMAGE DEHAZING

GAN-based image dehazing enhances quality in hazy conditions, but poor atmospheric conditions can degrade it. This paper discusses various dehazing methods, enhancing their practical applications and overall research utility.

i. Underwater image dehazing

GANs for underwater image dehazing[40] hold significant potential for improving accuracy and genuinely enhanced images. Beyond simply enhancing visuals, GANs can be applied to underwater image restoration for various purposes, including marine research, exploration, resource management, and automated tasks like simultaneous localization and Mapping (SLAM) in murky underwater environments. The MuLA-GAN has shown excellent performance in restoring essential image qualities like color, contrast, texture, and saturation in underwater images, proving their value for marine research and exploration.

ii. Remote sensing image dehazing

Satellite imagery [41] is essential for acquiring comprehensive data about the earth's surface, but atmospheric conditions such as haze frequently hinder its accuracy. To tackle the issue of haze in remote sensing images, different methodologies have been created to enhance image clarity and enhance the accuracy of interpretation.

iii. Intelligent transportation systems (ITS)

Image dehazing significantly contributes to improving visibility and detail in images used for autonomous navigation, traffic control, and security surveillance, especially under hazy conditions[42].

iv. Collection of Evidence

Enhancing surveillance image quality through dehazing can improve the reliability of evidence in crime investigations and traffic monitoring. By mitigating hardware limitations

and adverse atmospheric conditions, dehazing aids in suspect identification and law enforcement, making surveillance systems more effective [43].

CONCLUSION

This paper explores various image-dehazing methods from multiple perspectives, including methods, techniques, datasets, and performance evaluation across different metrics. By analyzing a range of studies, we highlight the merits and limitations of each approach. A key focus is a distinction between methods used before and after the introduction of GAN, highlighting a transformative shift in dehazing techniques. While GANs have demonstrated promising results, further advancements can be achieved by integrating GAN-based approaches with existing dehazing techniques. Moreover, no single method is universally effective for all weather conditions; each technique performs optimally under specific circumstances. As a result, researchers are increasingly exploring hybrid approaches that combine multiple methods. However, determining the optimal sequence for applying these techniques remains an active and widely debated research challenge. Future advancements should aim at reducing computation overhead, improving the capability for real-time image dehazing, and enhancing the robustness of the system against the variation of the real-world environment in process. Such transformations

could enhance the applicability of next-generation image dehazing models in various fields, from autonomous vehicle driving, medical imaging, and remote sensing to real-time surveillance systems.

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